Provenance Statement – Consent Before Code Briefing

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1. Purpose and scope

Every figure on the evidence slide (Slide 5) of the *Consent Before Code* briefing comes from one public source: the Australian Bureau of Statistics (ABS) release on Aboriginal and Torres Strait Islander language use from the 2021 Census (ABS, 2022a). This statement traces each figure from that source, through the data file and analysis script, to the published chart, so that a reader can reproduce or check it.

The statement covers four submitted files:

File	Role
`language_data.csv`	Native data file: ABS figures as published, with a source column
`make_chart.py`	Analysis and visualisation script (Python)
`figure_language_concentration.png`	Output chart (300 dpi), placed on Slide 5
`Consent Before Code Briefing.pptx`	Slide deck that presents and interprets the evidence

2. Data source

The data come from an official national statistical release, chosen because it is public, authoritative and counts people using each language at home.

Item	Detail
Publisher	Australian Bureau of Statistics (ABS)
Title	*Language Statistics for Aboriginal and Torres Strait Islander Peoples, 2021* (ABS, 2022a)
URL	https://www.abs.gov.au/statistics/people/aboriginal-and-torres-strait-islander-peoples/language-statistics-aboriginal-and-torres-strait-islander-peoples/2021
Date accessed	11 October 2026
Collection	2021 Census of Population and Housing; self-reported language used at home
Unit of analysis	Aggregate counts of people and languages; no individual-level or identifying data
Licence	Creative Commons Attribution 4.0 International (CC BY 4.0), the ABS's standard licence; reuse requires attribution, given on the chart and slides

A second ABS release (ABS, 2022b) and the *National Indigenous Languages Report* (DITRDC, 2020) are cited on Slide 4 for context only. No figures from them enter the data file or the chart.

3. Extraction: from the ABS release to the data file

Nine published figures were entered by hand into `language_data.csv`, exactly as the ABS reports them, with no rounding, weighting or adjustment. Each row records its source in a `source` column, so every value can be checked against the release.

Measure (CSV row)	Value	Used in chart?
People using an Indigenous language at home, 2021	76,978	Yes (Panel B denominator)
Share of Indigenous population, 2021	9.5%	No (cited on Slide 4)
People using an Indigenous language at home, 2016	63,754	No (context only)
Share of Indigenous population, 2016	9.8%	No (context only)
Languages actively used (lower bound)	151	Yes (Panel A)
Languages used by more than 1,000 people	19	Yes (Panel A)
Home users: Yumplatok (Torres Strait Creole)	7,596	Yes (Panel B)
Home users: Kriol	7,403	Yes (Panel B)
Home users: Djambarrpuyngu	3,839	Yes (Panel B)

One interpretive decision. The ABS states that "over 150" languages were actively used but gives no exact figure. The value 151 was entered as the smallest number consistent with that statement. This choice is recorded in the CSV's source column and in the chart's footnote. Because it is a lower bound, every share derived from it is conservative (see Section 6).

4. Processing and analysis

Every number on the chart is calculated by `make_chart.py` from the CSV at run time; none is typed into the chart by hand. The script runs in four steps:

1. **Load.** Reads `language_data.csv` into a measure-to-value dictionary using Python's standard `csv` module.
2. **Derive.** Calculates the figures below from the published values.
3. **Check.** An `assert` statement stops the script if the number of small languages falls below 132, guarding against a data-entry error.
4. **Report.** Prints the derived figures to the console so they can be compared with the chart and slide notes.

Derived figure	Calculation	Result
Languages with 1,000 or fewer home users	151 − 19	at least 132
Share of languages that are small	132 ÷ 151	at least 87%
Share of languages with more than 1,000 users	19 ÷ 151	at most 12.6%
Home users of the three largest languages	7,596 + 7,403 + 3,839	18,838 people
Top-three share of all home users	18,838 ÷ 76,978	24.5%

The script was run on 14 October 2026 and printed: "at least 132", "at most 12.6%" and "18,838 people = 24.5% of 76,978", matching the chart and the Slide 5 speaker notes.

5. Visualisation

The script produces one figure, `figure_language_concentration.png` (300 dpi), which appears unaltered on Slide 5.

- **Panel A** compares the 19 languages with more than 1,000 home users against the at least 132 with 1,000 or fewer.
- **Panel B** shows home users of the three most-used languages and annotates their combined 24.5% share.
- **Design choices.** Horizontal bars keep long labels readable. A single ochre highlight marks the key-message bar; all context bars are grey. Axes start at zero so bar lengths are not exaggerated.
- **Attribution.** The source (ABS, 2022a), the 151 lower-bound note and the caveat "Census counts home use, not fluency" are printed on the figure itself, so they travel with it if it is reused.

6. Exclusions, limitations and ethical handling

Excluded. Language counts from secondary websites were left out because they could not be verified against the ABS. Figures from the fourth National Indigenous Languages Survey (NILS4: 177 languages, 45 strong) are discussed on Slide 7 but not charted, because the Productivity Commission (2026) notes they are not comparable with earlier baselines.

Limitations of the data.

- *Home use is not fluency.* The Census records language used at home, not proficiency or custodial authority, so the number of people able to validate AI output may be smaller than shown.
- *Self-report and Census coverage.* Answers are self-reported, and the Census undercounts some remote communities.
- *Lower bound.* Using 151 for "over 150" means the 132 and 87% figures are minimums; the true share of small languages can only be higher.
- *Two language counts.* Slide 4 cites 167 languages reported in the Census, while the chart uses the narrower "over 150 actively used" measure. They are different ABS measures and are not combined.

Ethical handling. The data are aggregate, public and non-identifying, so no individual can be identified and no ethics approval was required to reuse them. The figures nonetheless describe Aboriginal and Torres Strait Islander peoples' cultural knowledge. They were handled in line with the CARE principles (Carroll et al., 2020) and the AIATSIS Code of Ethics (2020): they are used only to argue for community consent and benefit, are attributed to their source, and are not used to characterise or rank particular communities.

7. Tools and generative AI declaration

Tool	Version	Used for
Python	3.14 (64-bit)	Running the analysis script
matplotlib	**[insert version]**	Drawing the chart
Visual Studio Code	—	Editing and running the script
Microsoft PowerPoint	—	Building the slide deck

Generative AI use: [State which AI tools you used, if any, and for what (e.g. drafting code, checking calculations, wording this statement). State that you checked every figure against the ABS release yourself. If your unit requires a particular declaration wording, use it here.]

References

Australian Bureau of Statistics. (2022a). *Language statistics for Aboriginal and Torres Strait Islander peoples, 2021*. https://www.abs.gov.au/statistics/people/aboriginal-and-torres-strait-islander-peoples/language-statistics-aboriginal-and-torres-strait-islander-peoples/2021

Australian Bureau of Statistics. (2022b). *Aboriginal and Torres Strait Islander people: Census, 2021*. https://www.abs.gov.au/statistics/people/aboriginal-and-torres-strait-islander-peoples/aboriginal-and-torres-strait-islander-people-census/2021

Australian Institute of Aboriginal and Torres Strait Islander Studies. (2020). *AIATSIS code of ethics for Aboriginal and Torres Strait Islander research*. https://aiatsis.gov.au/sites/default/files/2022-02/aiatsis-code-ethics-jan22.pdf

Carroll, S. R., Garba, I., Figueroa-Rodríguez, O. L., Holbrook, J., Lovett, R., Materechera, S., Parsons, M., Raseroka, K., Rodriguez-Lonebear, D., Rowe, R., Sara, R., Walker, J. D., Anderson, J., & Hudson, M. (2020). The CARE principles for Indigenous data governance. *Data Science Journal*, *19*(1), 43. https://doi.org/10.5334/dsj-2020-043

Department of Infrastructure, Transport, Regional Development and Communications. (2020). *National Indigenous languages report*. https://www.arts.gov.au/what-we-do/indigenous-arts-and-languages/national-indigenous-languages-report

Productivity Commission. (2026). *Closing the Gap information repository: Socio-economic outcome area 16 dashboard*. https://www.pc.gov.au/closing-the-gap-data/dashboard/socioeconomic/outcome-area16